MID PROJECT

Congratulations, you are halfway there and it's time to put into practice what you've learned⚡

Project Overview:

- **Real-world data rarely comes clean. You will gather data from a variety of sources and in a variety of formats, assess its quality, then clean it. After that you will analyse a dataset and then communicate your findings using data visualization tools you have learned before. You will document your wrangling and analysis efforts in a Jupyter Notebook using Python (and its libraries).**

Dataset (optional):

- you can search for a dataset in any domain you interested in and prepare a proposal to be reviewed and find if it's suitable for this project (you can search for your dataset here)

  Note: The data you chose must be the same as the recommended data in terms of
  1- Size (Number of columns and rows).
  2- Quality (Don't choose a cleaned data because that's your mission).
  3- Simulates a real-time problem.
  4- Your proposal must show the description of your dataset and prove that it's not a clean one .
- Also , There are more than 10 datasets that we recommend for you, choose any of them and start your project! here
- **You will loss 5 grades by default if you didn't search for a dataset for your project and just pickup one from our dataset.**

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Project criteria (The standard by which we judge):

1- Code Functionality

Does the code work?	**The whole code must be functional and produces no errors when run.** **-The given code is sufficient to reproduce the results described.**
Does the project use good coding practices?	-The code makes use of functions to avoid repetitive code. -The code contains descriptive comments and variable names, making it easy to read.

2- Quality of Analysis

Is the question **clearly posed?**	**-The project clearly states one or more questions, then addresses those questions in the rest of the analysis.**

3- Data Cleaning Phase

Is the data cleaning well documented?	**-The project documents any changes that were made to clean the data, such as merging multiple files, handling missing values, etc.**

4- Exploration Phase

Is the data explored in many ways?	**-The project investigates the stated question(s) from multiple angles.** **-At least six variables are investigated using both Univariate Analysis (1d) and Bi/multi-variate Analysis .**
Does a variety of relevant visualizations and statistical summaries exist?	-The project's visualizations are varied and show multiple comparisons and trends. Relevant statistics are computed throughout the analysis when an inference is made about the data. -At least 5 kinds of plots should be included as part of the explorations.

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5- Visualization

Does the data visualize use appropriate plots and parameter choices?	**Visualizations made in the project depict the data in an appropriate manner that allows plots to be readily interpreted.**

Important Notes for each student for submission:

1- Must submit a notebook including all his work.
2- Must submit the dataset
3- Must submit a video that explain everything done in the project.
4- Must create a repo for his project and mention the main Epsilon AI repo.

Important Notes for each instructor for submission:

1- Must follow the grading template.
2- Must hold a discussion with students.

"It's your time to shine♡"

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